<u>Exp.</u> Consider $A=\begin{pmatrix}1&0\\0&1\end{pmatrix}$ and $B=\begin{pmatrix}1&1\\0&1\end{pmatrix}$.

We have $P_A(x)=P_B(x)=(1-x)^2=(x-1)^2$.

<u>Exp.</u> Consider $J_{\lambda,n}\in M_{n\times n}(K)$, $\lambda\in K$,

$$J:=J_{\lambda,n}:=\begin{pmatrix}\lambda&1&&&0\\&\lambda&1&&\\&&\lambda&1&\\0&&&\ddots&1\\&&&&\lambda\end{pmatrix}$$ (Jordan block)

$\leftarrow$ λ's along the diag. 1's along "diag" above the diag of the λ's, 0 everywhere else

$P_{J_{\lambda,n}}(x)=(\lambda-x)^n$. $\Rightarrow$ J has only one eigenvalue, λ.

<u>Eigenvectors for the eigenvalue λ:</u>

$$J-\lambda\cdot I_n=\begin{pmatrix}0&1&&&0\\&0&1&&\\&&\ddots&\ddots&\\&&&\ddots&1\\0&&&&0\end{pmatrix}$$

$\Rightarrow \operatorname{rank}(J-\lambda\cdot I_n)=n-1$

$\Rightarrow \operatorname{Ker}(J-\lambda I_n)=1$.

$\Rightarrow \operatorname{Eig}_J(\lambda)$ is 1-dim.

$(n-1)$ lin. indepnd. cols

Clearly $e_1=\begin{pmatrix}1\\0\\\vdots\\0\end{pmatrix}$ is an eigenvect. of λ.

$\Rightarrow \operatorname{Eig}_J(\lambda)=\operatorname{Sp}(e_1)$.

Important remark about the difference between polynomials $P(x)\in K[x]$ and the functions the define $K\to K$.

For example, if $K=\mathbb{Z}_2$ then $p(x)=x$ and $q(x)=x^2$ are <u>different</u> polynomials but they define the same funct. $\mathbb{Z}_2\to\mathbb{Z}_2$. (What happens for $K=\mathbb{Q},\mathbb{R},\mathbb{C},\dots$?)

<u>Lemma.</u> **(21.b.1)** Let $A \in M_{n\times n}(K)$ (or $T \in \mathrm{End}(V)$) be diagonalizable. Then

$P_A(x)$ decomposes as a product of linear (i.e. deg 1) factors

$P_A(x) = (\lambda_1 - x)\cdot \ldots \cdot (\lambda_n - x)$, where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A (but there might be repetitions among the λ_i's).

<u>Proof.</u> By assumption $\exists$ an invertible mat. P s.t.

$$P^{-1} A P = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \leftarrow \Lambda \text{ denote by}$$

(i.e. A is similar to Λ.)

$$\Rightarrow P_A(x) = P_\Lambda(x) = \det \begin{pmatrix} \lambda_1 - x & & 0 \\ & \ddots & \\ 0 & & \lambda_n - x \end{pmatrix} = (\lambda_1 - x)\cdot \ldots \cdot (\lambda_n - x). \quad \blacksquare$$

<u>Exp.</u> $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $K = \mathbb{R}$ ($T_A : \mathbb{R}^2 \to \mathbb{R}^2$ is rot. by $90°$ in the counter clockwise direction.)

$P_A(x) = \det \begin{pmatrix} -x & -1 \\ 1 & -x \end{pmatrix} = x^2 + 1$. This polynom. has no roots in $\mathbb{R}$. And of course $\nexists$ eigenvalues.

A quick digression about polynomials.

Let $0 \neq p(x) \in K[x]$, $\lambda \in K$. By dividing $p(x)$ by $(x-\lambda)$ we can write $p(x) = (x-\lambda)\cdot q(x) + R$ (reminder) where $\deg q(x) = \deg p(x) - 1$ and $R \in K$. But if we substitute $x = \lambda$ we obtain

$$p(\lambda) = 0 \cdot q(\lambda) + R = R. \Rightarrow p(x) = (x-\lambda)\cdot q(x) + p(\lambda).$$

Conclusion. $p(\lambda) = 0 \iff p(x) = (x-\lambda)\cdot q(x)$ for some $q(x) \in K[x]$.

We say that λ is a zero of $p(x)$ (De: Nullstelle) or a root of $p(x)$.

Sometimes λ is also a zero of $q(x)$, so $q(x) = (x-\lambda)\cdot h(x)$

$\Rightarrow p(x) = (x-\lambda)^2 \cdot h(x)$, etc.

Def. (21.b.2) Let $0 \neq p(x) \in K[x]$, $\lambda \in K$. We say that λ is a zero of multiplicity $m \in \mathbb{N}_{>0}$ if $p(x) = (x-\lambda)^m \cdot g(x)$, where $g(x) \in K[x]$ and $g(\lambda) \neq 0$. ($\iff \lambda$ is not a zero of $g(x)$.)

If $m \geq 2$ we say λ is a multiple zero of $p(x)$.

If $m = 1$ we say λ is a simple zero of $p(x)$.

Geometric & algebraic multiplicities of eigenvalues.

Def. (21.b.3) Let V be a vect. sp. over K and $T \in \mathrm{End}(V)$. Let $\lambda \in K$ be an eigenvalue of T. We define the geometric multiplicity of λ to be $m_g(T;\lambda) := \dim \mathrm{Eig}_T(\lambda)$.

Assume now, in addition, that V is f.dim. We define the algebraic multiplicity of λ to be the multiplicity of the zero λ in the charact. polynom. $P_T(x)$ of T. We denote it by $m_a(T;\lambda)$.

Prop. (21.b.4) $m_g(T;\lambda) \leq m_a(T;\lambda)$.

Proof. Let $v_1, \dots, v_k$ be a basis for $\mathrm{Eig}_T(\lambda)$. We extend this basis to a basis $\mathcal{B} = (v_1, \dots, v_k, w_{k+1}, \dots, w_n)$ of V.

Then we have $[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} | & & | & | & & | \\ [Tv_1]_{\mathcal{B}} & \cdots & [Tv_k]_{\mathcal{B}} & [Tw_{k+1}]_{\mathcal{B}} & \cdots & [Tw_n]_{\mathcal{B}} \\ | & & | & | & & | \end{pmatrix} =$

$$= \begin{array}{c} \\ k \updownarrow \\ \\ \end{array} \begin{array}{c} \overset{k}{\longleftrightarrow} \\ \left(\begin{array}{ccc|c} \lambda & & 0 & \\ & \ddots & & * \\ 0 & & \lambda & \\ \hline & 0 & & C \end{array}\right) \\ \underset{n-k}{\longleftrightarrow} \end{array} \begin{array}{c} \\ \\ \\ n-k \updownarrow \end{array}, \quad \text{where } C \in M_{(n-k)\times(n-k)}(K)$$

$\Rightarrow P_T(x) = \det\left([T]_{\mathcal{B}}^{\mathcal{B}} - x \cdot I_n\right) = \dots = (\lambda - x)^n \cdot \det(C - xI_{n-k}) =$
$= (\lambda - x)^k \cdot P_C(x) = (-1)^k (x-\lambda)^k \cdot P_C(x)$. $\Rightarrow$ the multip. of λ as a zero of $P_T(x)$ is a least k.

$\Rightarrow m_a(T;\lambda) \geq k = m_g(T;\lambda)$. ∎

<u>Exp.</u> Consider $J := J_{\lambda,n} := \begin{pmatrix} \lambda & 1 & & & 0 \\ & \lambda & 1 & & \\ & & \lambda & 1 & \\ & & & \ddots & 1 \\ 0 & & & & \lambda \end{pmatrix}$.

We saw that $p_J(x) = (\lambda - x)^n \Rightarrow m_a(J;\lambda) = n$.

—— " —— $\mathrm{Eig}_J(\lambda) =$ 1-dim sp. $\Rightarrow m_g(J;\lambda) = 1$.